At the Fefferman conference

Terence Tao · 6 May 2009

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I am currently at Princeton for the conference "The power of Analysis" honouring Charlie Fefferman's 60th birthday. I myself gave a talk at this conference entitled "Recent progress on the Kakeya conjecture"; I plan to post a version of this talk on this blog shortly.

But one nice thing about attending these sorts of conferences is that one can also learn some neat mathematical facts, and I wanted to show two such small gems here; neither is particularly deep, but I found both of them cute. The first one, which I learned from my former student Soonsik Kwon, is a unified way to view the mean, median, and mode of a probability distribution μ on the real line. If one assumes that this is a continuous distribution $\mu = f(x)\ dx$ for some smooth, rapidly decreasing function $f : \mathbb{R} \to \mathbb{R}^+$ with $\int_{\mathbb{R}} f(x)\ dx = 1$, then the mean is the value of x_0 that minimises the second moment

$$\int_{\mathbb{R}} |x - x_0|^2 f(x)\ dx,$$

the median is the value of x_0 that minimises the first moment

$$\int_{\mathbb{R}} |\ x - x_0|\ f(x)\ dx,$$

and the mode is the value of x_0 that *maximises* the "pseudo-negative first moment"

$$\int_{\mathbb{R}} \delta(x - x_0) f(x)\ dx.$$

(Note that the Dirac delta function $\delta(x - x_0)$ has the same scaling as $|x - x_0|^{-1}$, hence my terminology "pseudo-negative first moment".)

The other fact, which I learned from my former classmate Diego Córdoba (and used in a joint paper of Diego with Antonio Córdoba), is a pointwise inequality

$$|\nabla|^\alpha (f^2)(x) \leq 2 f(x) |\nabla|^\alpha f(x)$$

for the fractional differentiation operators $|\nabla|^\alpha$ applied to a sufficiently nice real-valued function $f : \mathbb{R}^d \to \mathbb{R}$ (e.g. Schwartz class will do), in any dimension d and for any $0 \leq \alpha \leq 1$; this should be compared with the product rule $\nabla(f^2) = 2f\nabla f$.

The proof is as follows. By a limiting argument we may assume that $0 < \alpha < 1$. In this case, there is a formula

$$|\nabla|^\alpha f(x) = c(\alpha) \int_{\mathbb{R}^d} \frac{f(x) - f(y)}{|x - y|^{d+\alpha}}\ dy$$

for some explicit constant $c(\alpha) > 0$ (this can be seen by computations similar to those in my recent lecture notes on distributions, or by analytically continuing such computations; see also Stein's "Singular integrals and differentiability properties of functions"). Using this formula, one soon sees that

$$2f(x)|\nabla|^\alpha f(x) - |\nabla|^\alpha (f^2)(x) = c(\alpha) \int_{\mathbb{R}^d} \frac{|f(x) - f(y)|^2}{|x - y|^{d+\alpha}}\ dy$$

and the claim follows.

Comments (3)



Kevin · 6 May 2009

Another way to view the mode, which is closer in form to the first two examples, is to think of it as the value of x_0 which minimizes the “0th” moment. That is, you look at the limiting case as the exponent on |x-x_0| goes to zero.

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[…] Math and more math. […]



J. · 7 May 2009

Hi — a very small detail — I think that last integral should be over R^d, not from R to d.
Just fyi! Thanks.
[Corrected, thanks – T]